

DP 024 (SET 1)
Physics 2
Semester II
Session 2024/2025
2 hours

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Fizik
Semester II
Sesi 2024/2025
2 jam



SES II
UNIT FIZIK
KOLEJ MATRIKULASI SELANGOR

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
PRE MATRICULATION PROGRAMME EXAMINATION

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DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

QUESTION	MARKS	SCORE
1	16	
2	12	
3	10	
4	14	
5	14	
6	14	
	80	

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Hydrogen mass	m_H	$= 1.673 \times 10^{-27} \text{ kg}$ $= 1.007825 \text{ u}$
Deuteron mass	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law	$k = \frac{1}{4\pi\epsilon_o}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE

- | | | | |
|-----|---|-----|---|
| 1. | $y = A \sin \omega t$ | 20. | $\tau = RC$ |
| 2. | $\omega = \frac{2\pi}{T} = 2\pi f$ | 21. | $Q = Q_o e^{\frac{-t}{RC}}$ |
| 3. | $v = A\omega \cos \omega t$ | 22. | $Q = Q_o \left(1 - e^{\frac{-t}{RC}}\right)$ |
| 4. | $a = -A\omega^2 \sin \omega t$ | 23. | $B = \frac{\mu_o I}{2\pi r}$ |
| 5. | $k = \frac{2\pi}{\lambda}$ | 24. | $B = \frac{\mu_o NI}{L}$ |
| 6. | $v = f\lambda$ | 25. | $B = \frac{\mu_o I}{2r}$ |
| 7. | $y(x, t) = A \sin(\omega t \pm kx)$ | 26. | $\phi = BA \cos \theta$ |
| 8. | $v = \frac{dy}{dt}$ | 27. | $\Phi = N \phi$ |
| 9. | $y = 2A \cos kx \sin \omega t$ | 28. | $\varepsilon = -\frac{d\phi}{dt}$ |
| 10. | $v = \sqrt{\frac{T}{\mu}}$ | 29. | $\varepsilon = Blv \sin \theta$ |
| 11. | $I = \frac{dQ}{dt}$ | 30. | $\varepsilon = -NA \frac{dB}{dt}$ |
| 12. | $Q = ne$ | 31. | $\varepsilon = -NB \frac{dA}{dt}$ |
| 13. | $\rho = \frac{RA}{l}$ | 32. | $R = 2f$ |
| 14. | $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 33. | $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ |
| 15. | $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 34. | $M = \frac{h_i}{h_o} = -\frac{v}{u}$ |
| 16. | $V = IR$ | 35. | $y_m = \frac{m\lambda D}{d}$ |
| 17. | $C = \frac{Q}{V}$ | 36. | $y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$ |
| 18. | $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ | 37. | $\Delta y = \frac{\lambda D}{d}$ |
| 19. | $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ | | |

1. (a) The vertical position, y of a particle along x -axis under periodic vibration is described by the equation

$$y(x, t) = 13 \sin(25\pi t - 0.2\pi x)$$

where y in cm, x in m and t in second. Determine

- (i) the amplitude.
- (ii) the direction of the wave propagation.
- (iii) the wavelength.
- (iv) the period.

[6 marks]

- (b) Two waves that undergo superposition are represented by the following equations:

$$y = 10.0 \cos x \sin 3t$$

where x and y are in cm and t is in second. Calculate

- (i) the distance between adjacent node and antinode in the standing wave.
- (ii) the speed of progressive wave.
- (iii) the amplitude of the particle vibrating at $x = 1.5$ cm.

[8 marks]

- (c) The tension in an 85 cm guitar string of mass 1.0 g is 64 N. When the string is plucked, calculate the velocity of the transverse wave on the string.

[2 marks]

2. (a) A wire 5.0 m long and 3.0 mm in diameter has a resistance of 100Ω . A 15 V of potential difference is applied across the wire. Determine

- (i) the current in the wire.
- (ii) the resistivity of the wire.

[5 marks]

- (b)

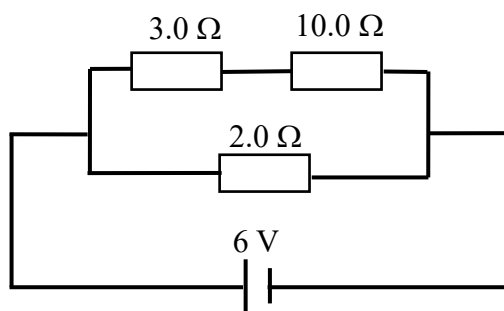


FIGURE 1

FIGURE 1 shows an arrangement of three resistors. For the circuit, calculate the

- (i) effective resistance.
- (ii) current that passes through 10.0Ω resistor.
- (iii) potential difference across 3.0Ω resistor.

[7 marks]

3. (a)

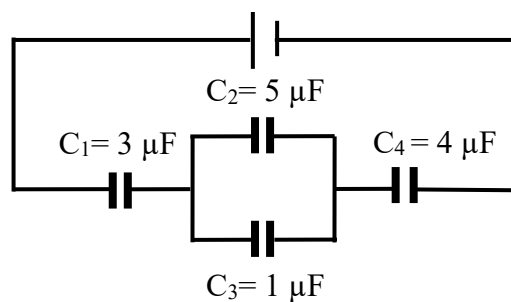
**FIGURE 2**

FIGURE 2 shows an arrangement of four capacitors.

- (i) Calculate the effective capacitance.
- (ii) If the potential difference across the $4\ \mu\text{F}$ capacitor is $5\ \text{V}$, calculate the potential difference across C_1 .

[5 marks]

(b) A $460\ \mu\text{F}$ capacitor is discharge through a $10\ \text{k}\Omega$ resistor. Calculate the

- (i) time constant.
- (ii) time taken for the charge to decrease 20% of the initial value.

[5 marks]

4. (a)

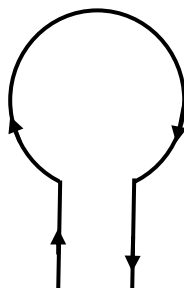
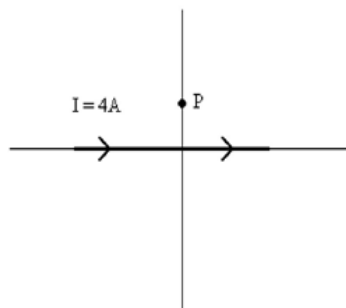
**FIGURE 3**

FIGURE 3 shows that current flows in clockwise direction through a circular coil. Copy the diagram and show the direction of magnetic field at the center of the circular coil.

[1 mark]

(b)

**FIGURE 4**

A long straight wire carrying a 4 A current is placed along the x axis as shown in **FIGURE 4**. Determine

- the magnitude of the magnetic field at a point P, located $y = 0.03$ m, due to the current in this wire.
- the direction of the magnetic field at a point P, located $y = 0.03$ m, due to this wire.

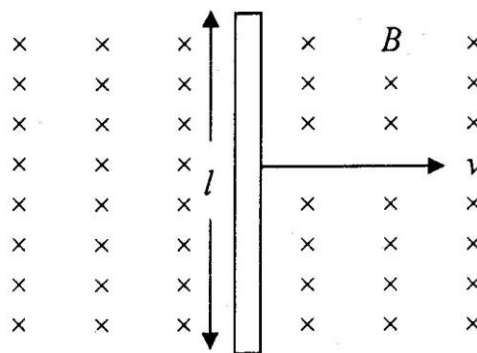
[3 marks]

- (b) A coil of wire consists of 20 turns. Each turn is a square of side, $d = 8$ cm. A uniform magnetic field directed perpendicular to the plane.

- If the induced emf generated in coil is 5 V, determine the rate of decreasing of magnetic field.
- Determine the total resistance in the coil if the induced current is 0.85 A.

[5 marks]

(c)

**FIGURE 5**

A bar of length $l = 0.2$ m moves to the right with a uniform velocity, $v = 0.5$ m s⁻¹ as indicated in **FIGURE 5**. If the resistance of the bar is $10\ \Omega$ and the magnetic field $B = 4$ T,

- calculate the induced emf in the bar.
- calculate the induced current.
- determine the direction of the induced current.

[5 marks]

5. (a)

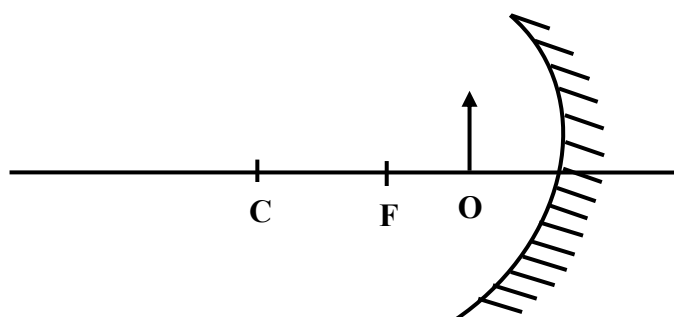
**FIGURE 6**

FIGURE 6 shows an object O placed in front of a concave mirror where C and F are centre of curvature and focal point.

- (i) Copy **FIGURE 6** and construct the ray diagram for formation of the image.
- (ii) State two (2) characteristics of the image.

[5 marks]

- (b) An object is placed 12 cm in front of a mirror. The real image formed is the same size as the object.

- (i) Determine the focal length of the mirror.
- (ii) State the type of mirror.

[4 marks]

- (c) A slide projector is required to produce a real image with height of 216 mm of an object 36 mm high. If the distance of the object from the lens is 0.1 m, calculate

- (i) the distance of the lens from the image.
- (ii) the focal length of the lens.

[5 marks]

- 6. (a) Young's double-slit experiment is performed using light of wavelength 570 nm. The distance between the slits and the screen is 2 m. The tenth dark fringe is observed 12.5 mm from the central maximum. Calculate

- (i) the slit separation.
- (ii) the distance of the third bright from the central maximum.
- (iii) What will happen to the size of fringes if the distance between the slits is increased? State your reason.

[8 marks]

- (b) In a double slit experiment the distance between slits is $d = 5.0$ mm and the distance to the screen is $D = 1.0$ m. There are two interference patterns on the screen, one due to violet light with $\lambda = 440$ nm and another due to red light with $\lambda = 630$ nm. Determine the separation distance from central bright, between fifth order of bright violet fringe and third order of dark red fringe.

[6 marks]